Port Scanning - Wireshark

What is a Port Scan

Port scanning is a network reconnaissance technique that identifies which ports are open on a computer or network device. This process helps determine the services running on the system, as certain applications listen on specific ports. For instance, HTTP uses port 80, DNS uses port 53, and SSH uses port 22.

Targeted IP - 192.168.80.130

Commands Used and Output

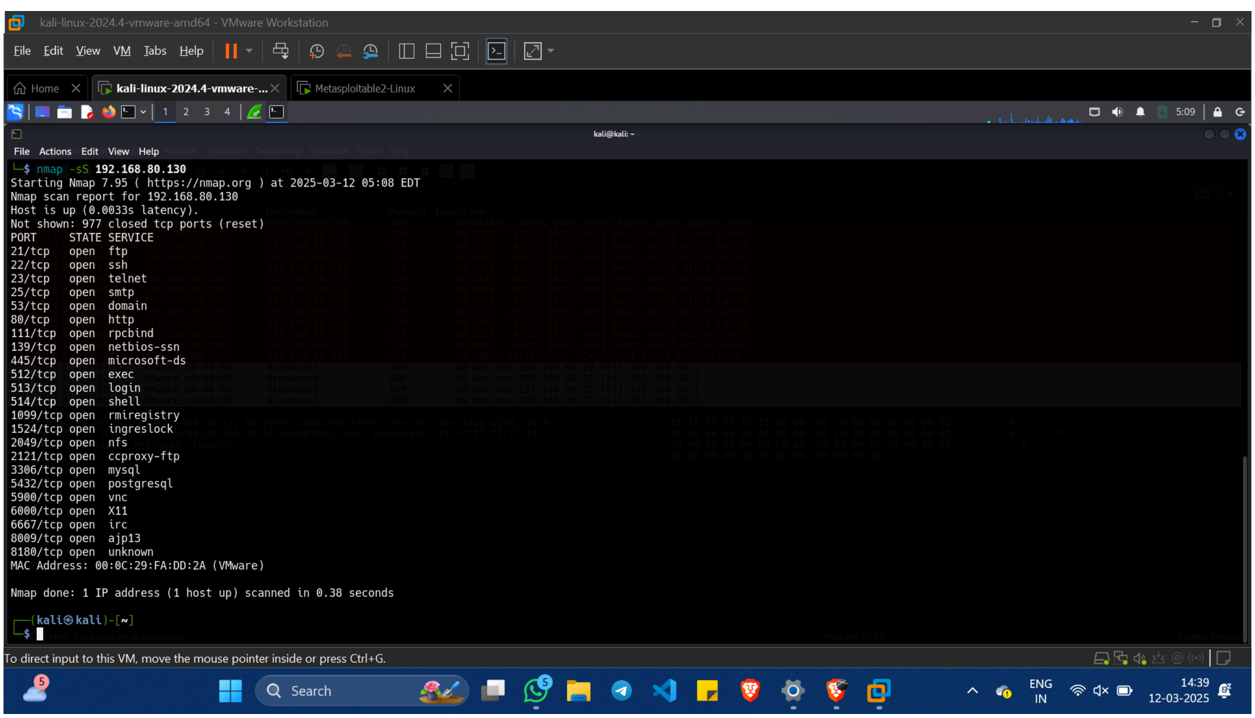
Stealth Scan (-sS)

This half-open scan sends SYN packets but does not complete the three-way handshake.

- If a SYN-ACK is received, the port is open.
- If an RST is received, the port is closed.
- This method is stealthier because it does not log a full connection.

- Sends SYN packets but does not complete the handshake.
  - SYN-ACK = Open port.
  - RST = Closed port.

Command: nmap -sS 192.168.80.130

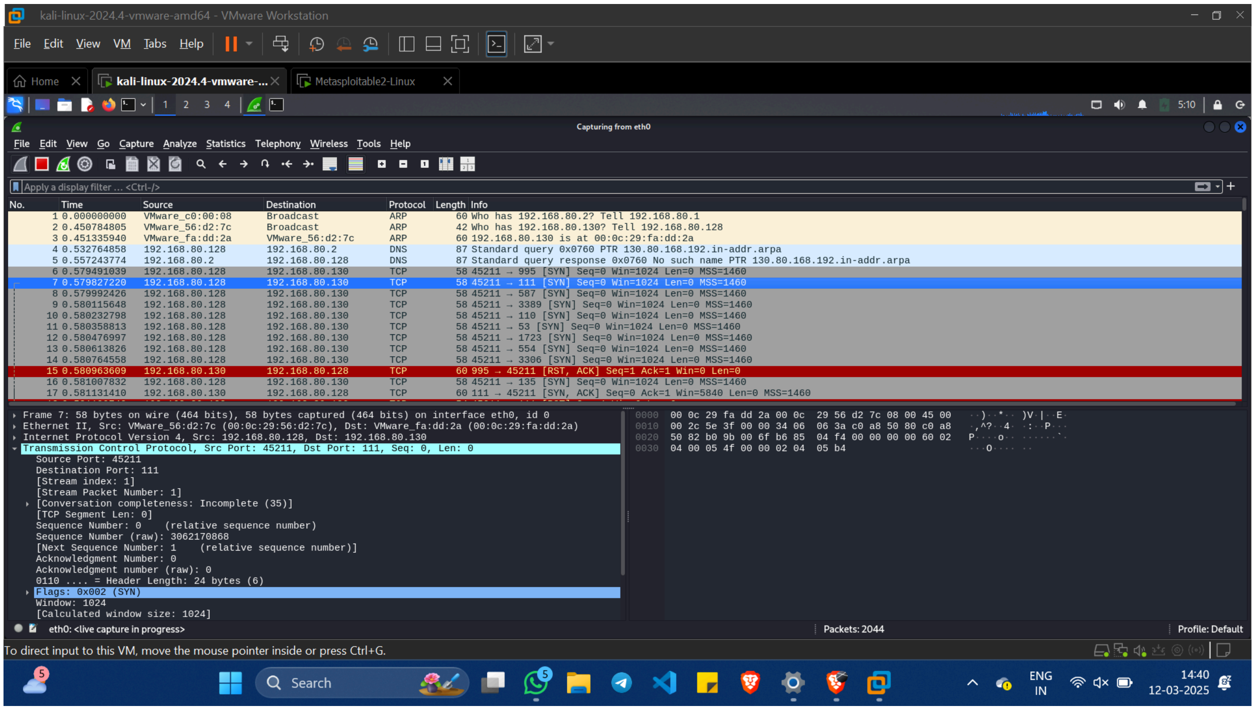


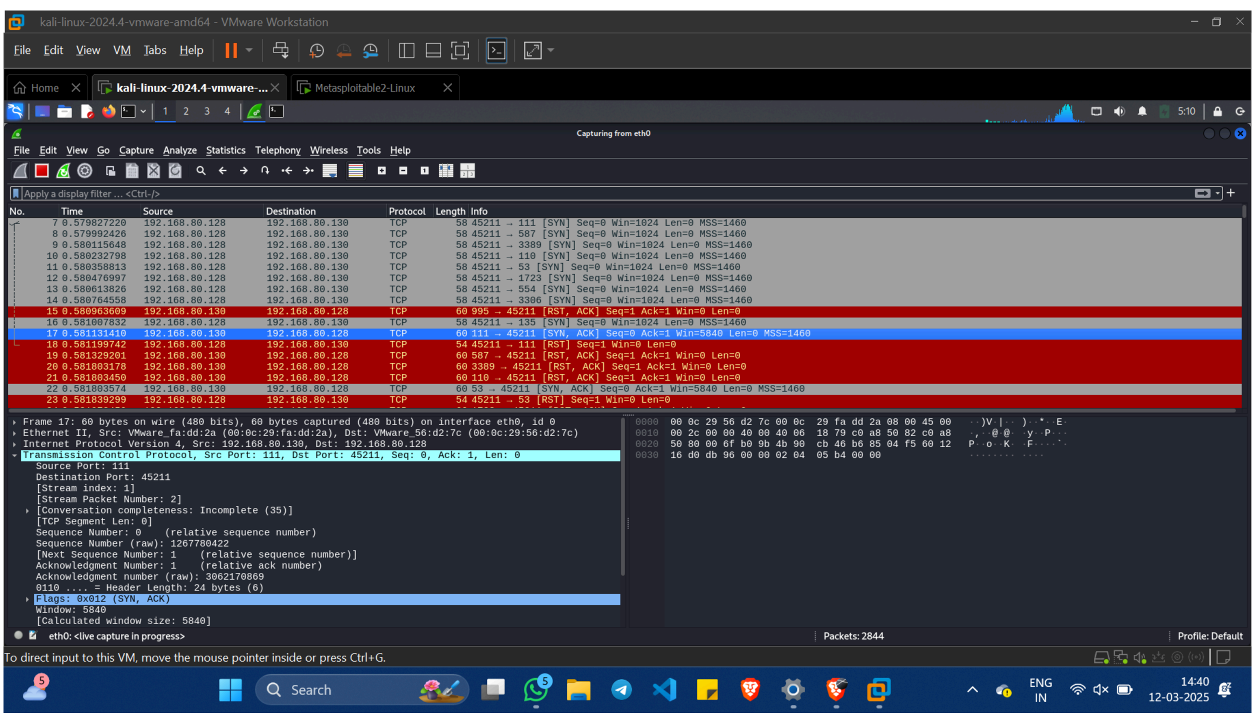
*Submitted by **Nanda Krishnan V**, MSc Cybersecurity*

Analysis:

- This scan checks open ports without completing the handshake.
- Found port 111 open.

Wireshark Analysis:





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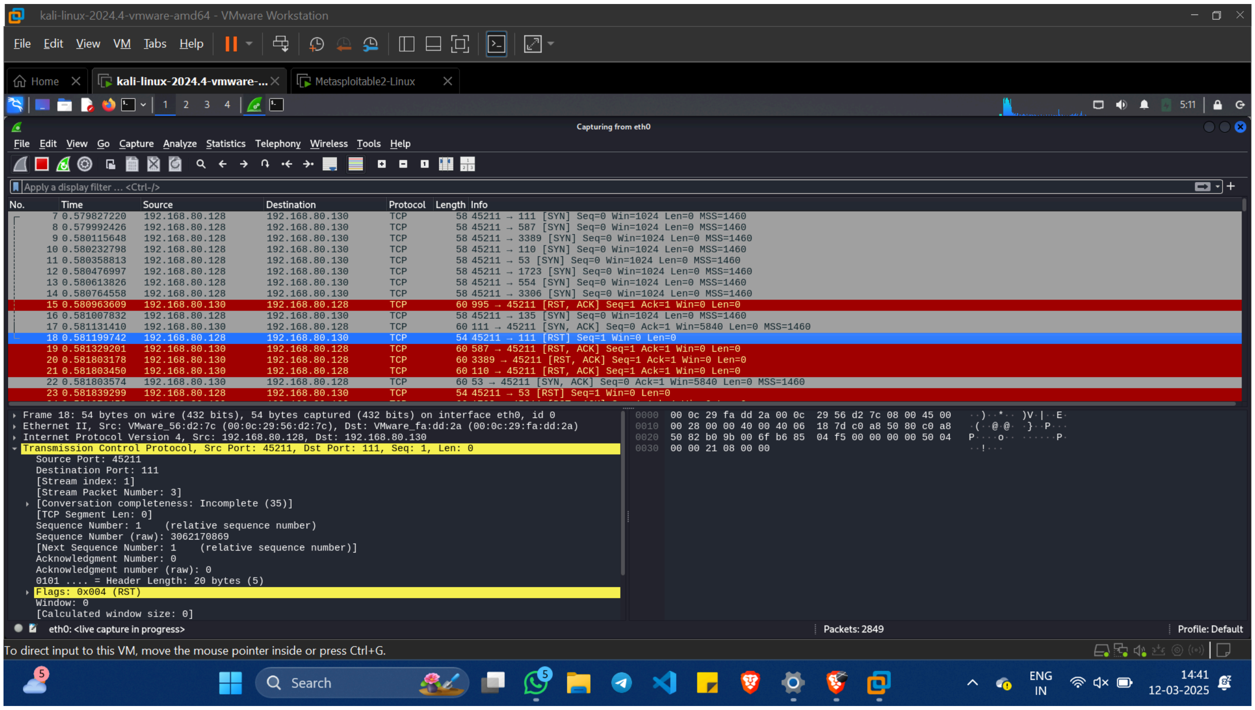
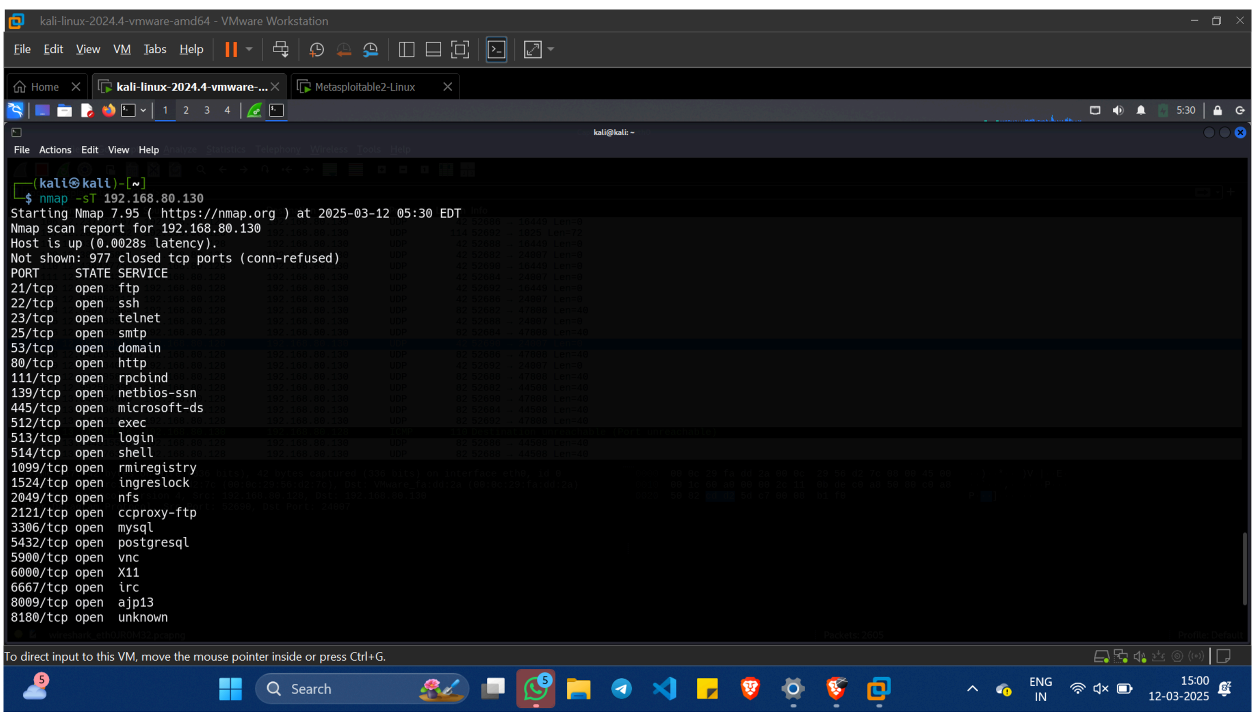


TCP Connect Scan (-sT)

Command: nmap -sT 192.168.80.130



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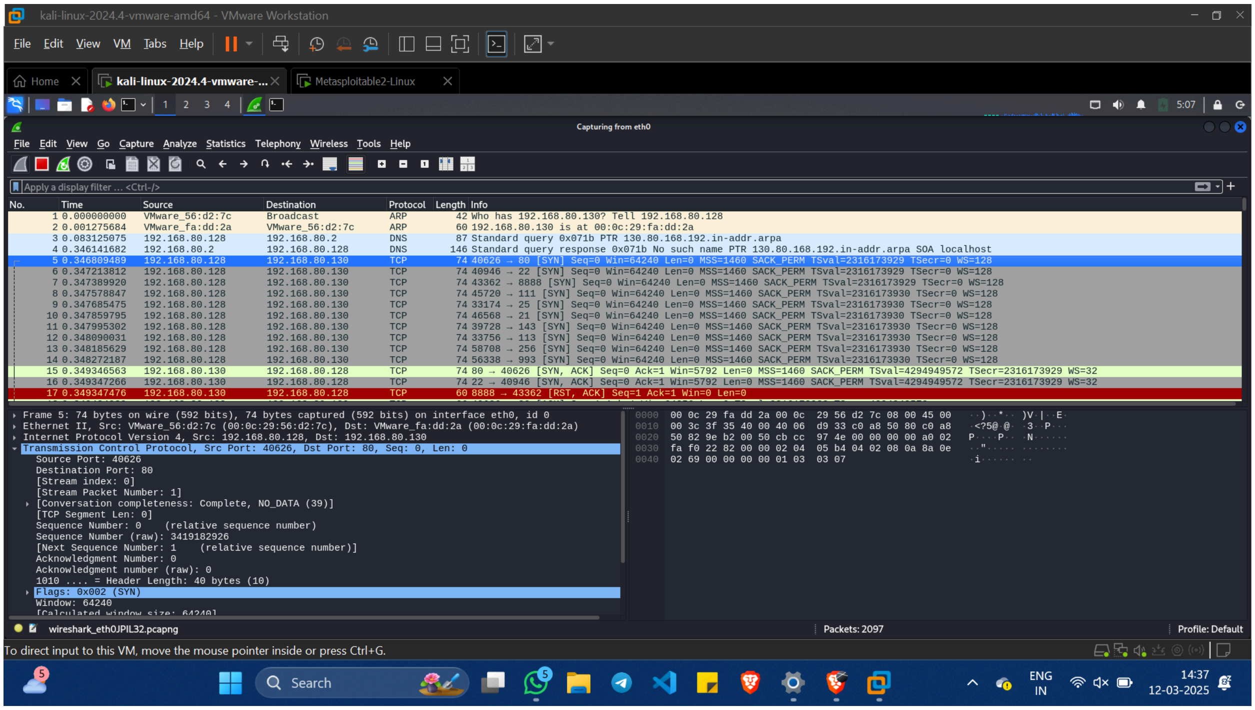
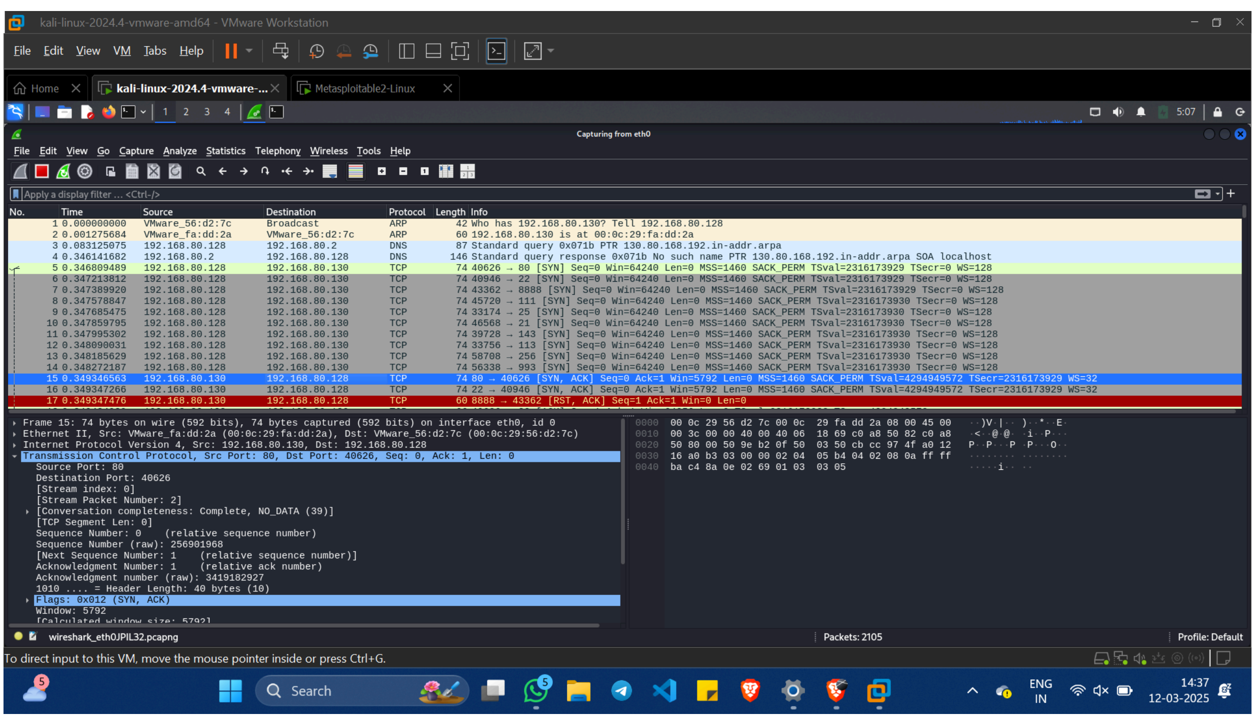
SYN: The client sends a SYN (synchronize) packet to the server, requesting a connection.



SYN-ACK: The server responds with a SYN-ACK (synchronize-acknowledge) packet, indicating it is ready.



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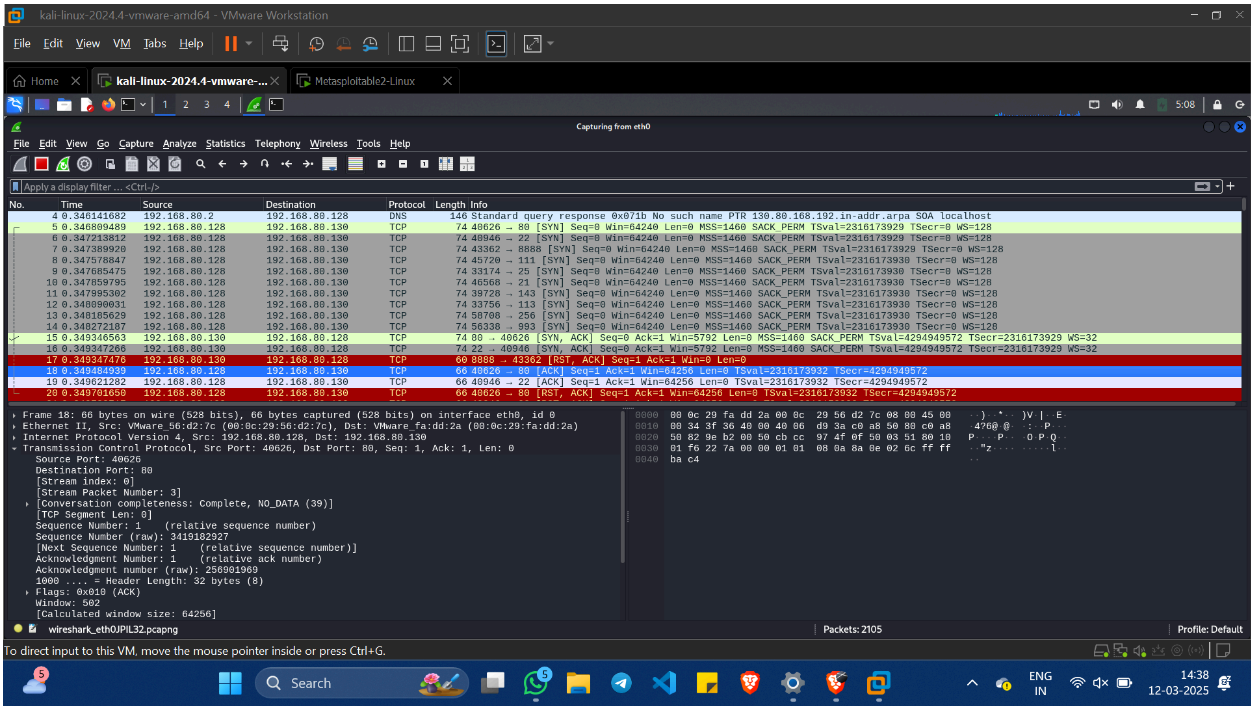
ACK: The client sends an ACK (acknowledge) packet to complete the handshake, establishing the connection.



Xmas Scan (-sX)

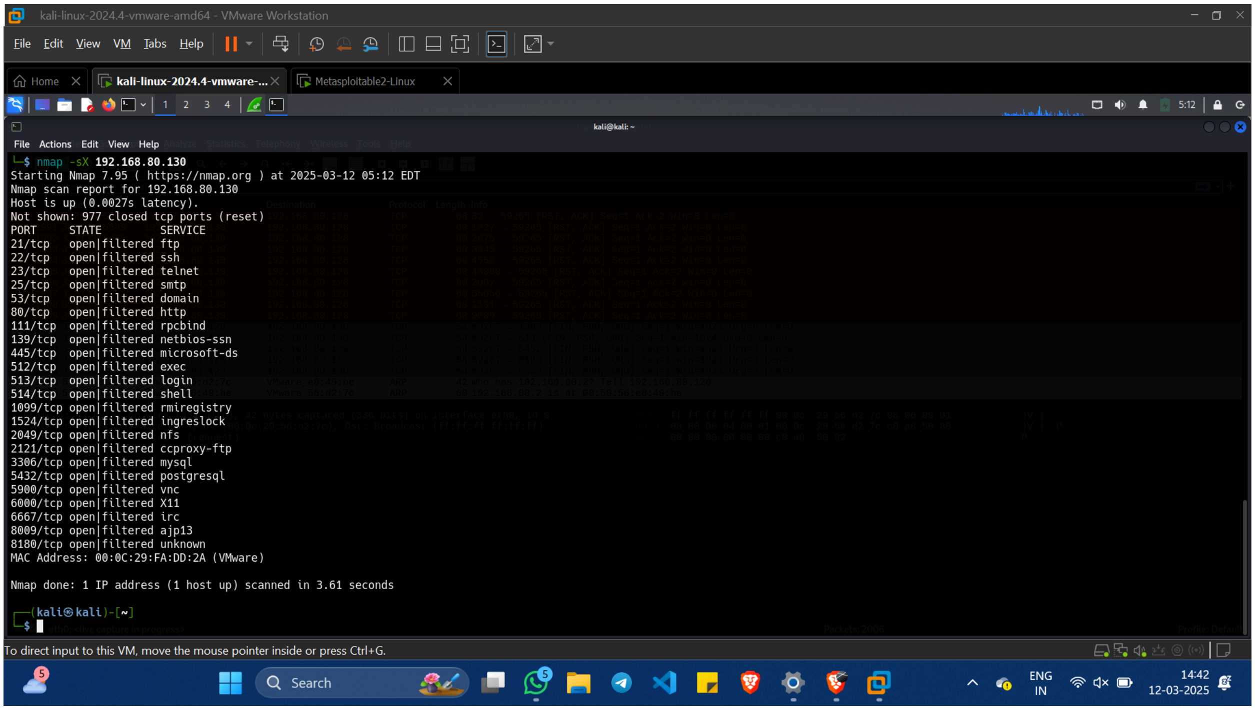
An Xmas scan, also known as a Christmas tree scan, is a type of port scanning technique used in computer security. It manipulates the flags in the TCP header to cause varied responses, aiding in the scanning process. The name "Xmas scan" refers to a packet configured with the FIN, URG, and PSH flags set, resembling a lit Christmas tree.

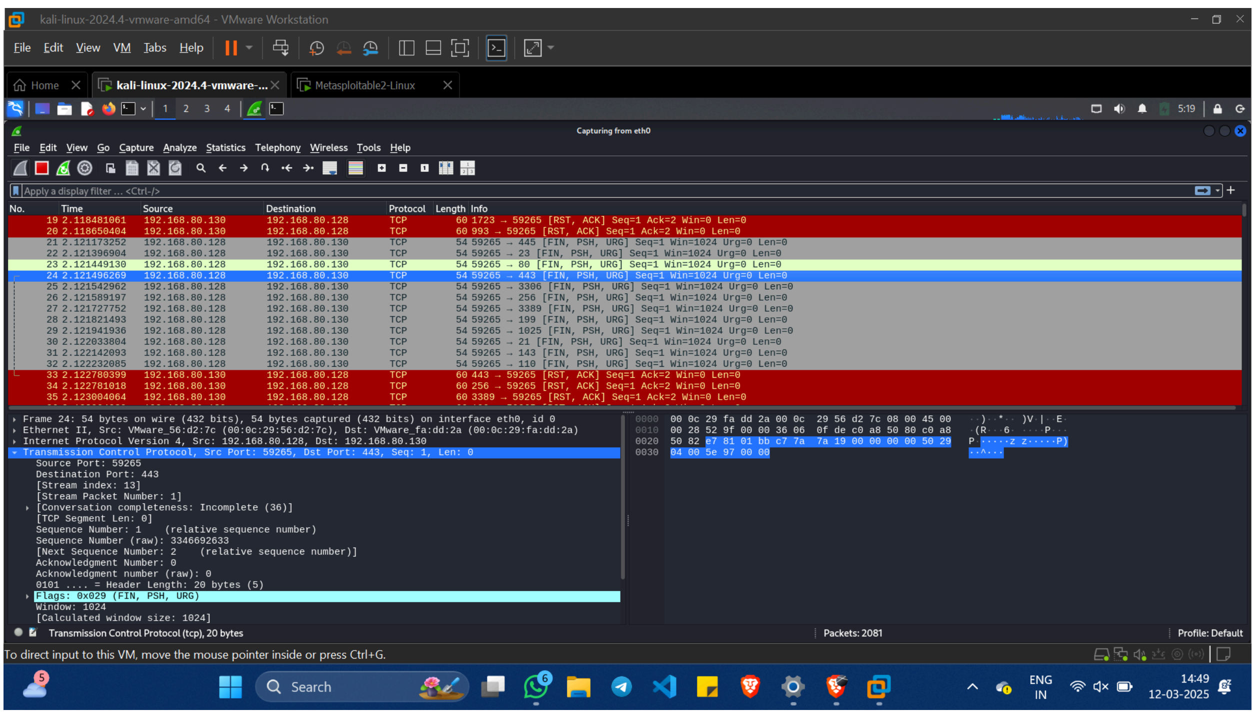
This scan sends packets with FIN, URG, and PUSH flags set, resembling a Christmas tree.

- If there is no response, the port is open or filtered.
- If an RST is received, the port is closed.

- Sends packets with FIN, URG, and PUSH flags.
  - No response = Open or filtered port.
  - RST response = Closed port.

Command: nmap -sX 192.168.80.130





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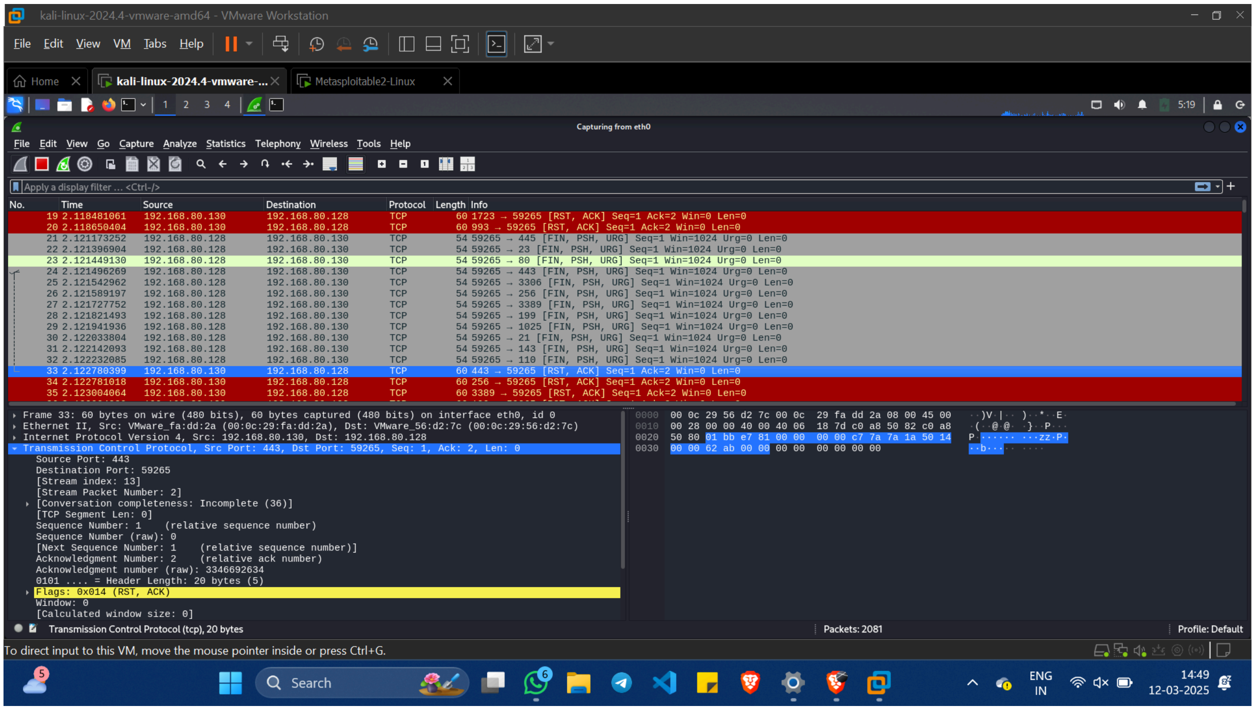


UDP Scan (-sU)

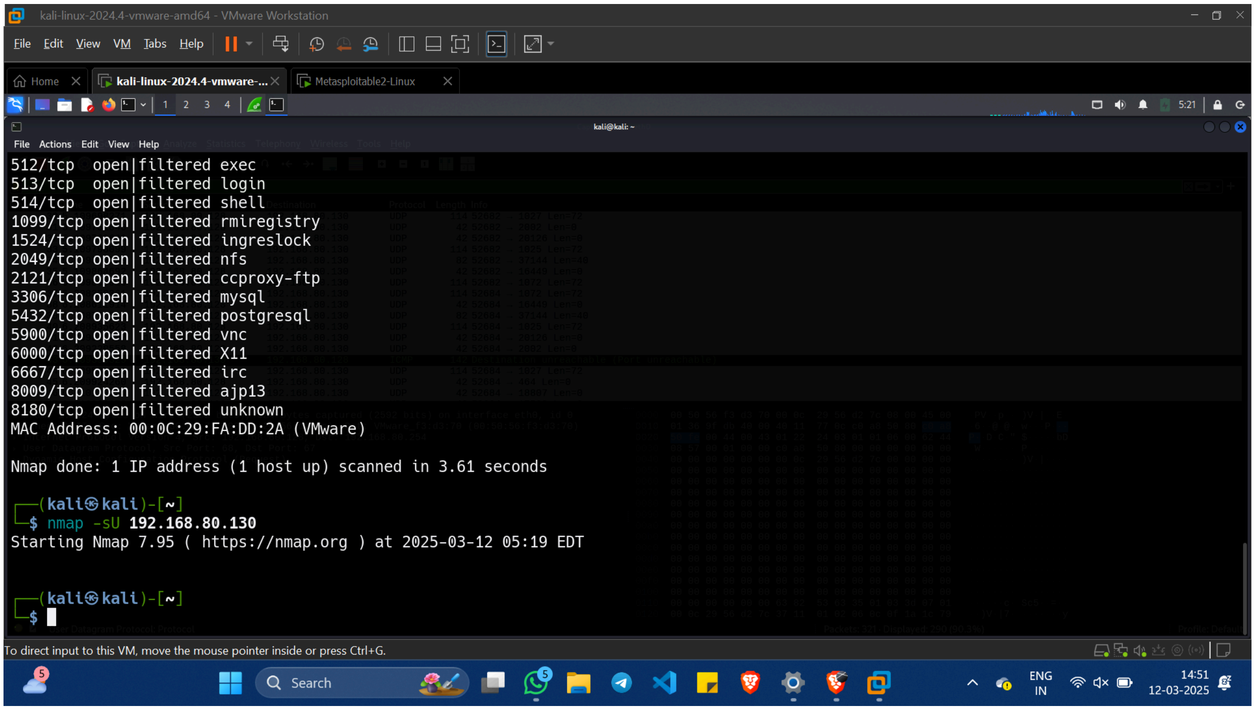
A UDP scan is a process used to identify open UDP ports on a target system or to discover UDP services that are currently running. Unlike TCP, which is a connection-oriented protocol, UDP is connectionless, making it harder to probe for open ports.

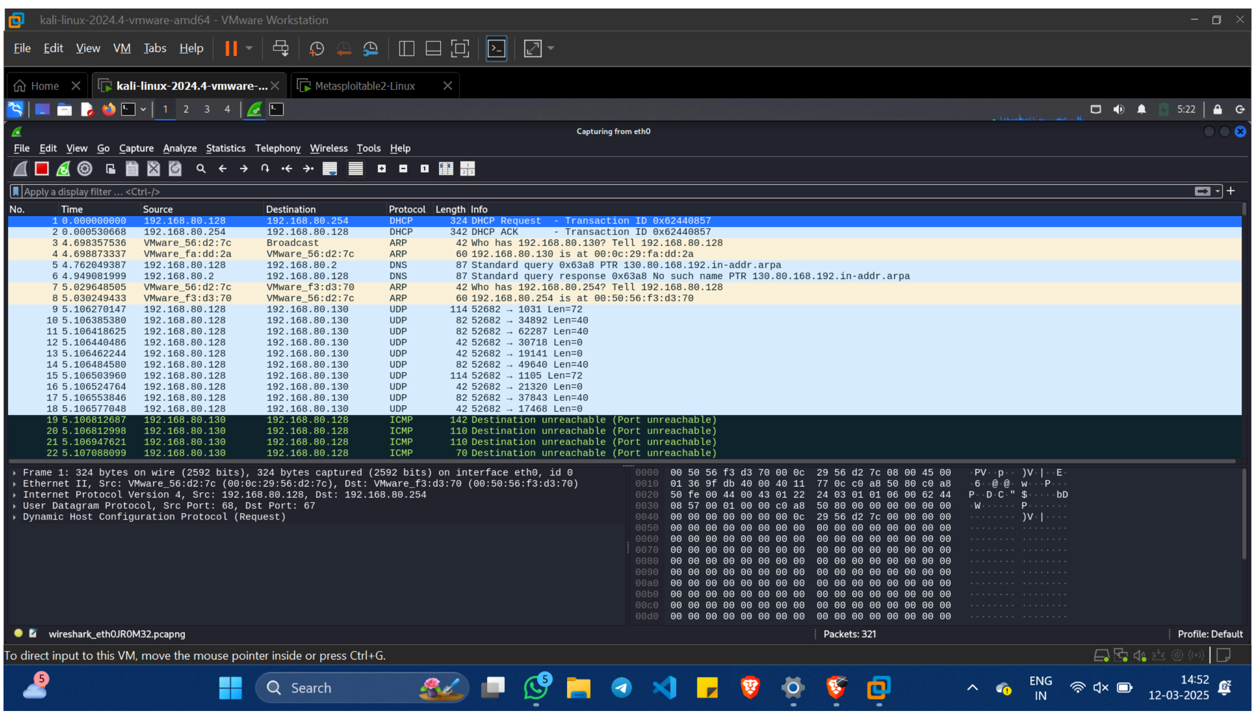
This scan sends UDP packets to detect open UDP ports.

- If a response is received, the port is open.
- If there is no response, the port may be filtered.
- If an ICMP Port Unreachable message is received, the port is closed.

- Sends UDP packets to find open UDP ports.
    - Response = Open port.
    - No response = Possibly filtered.
    - ICMP Port Unreachable message = Closed port.

Command: nmap -sU 192.168.80.130





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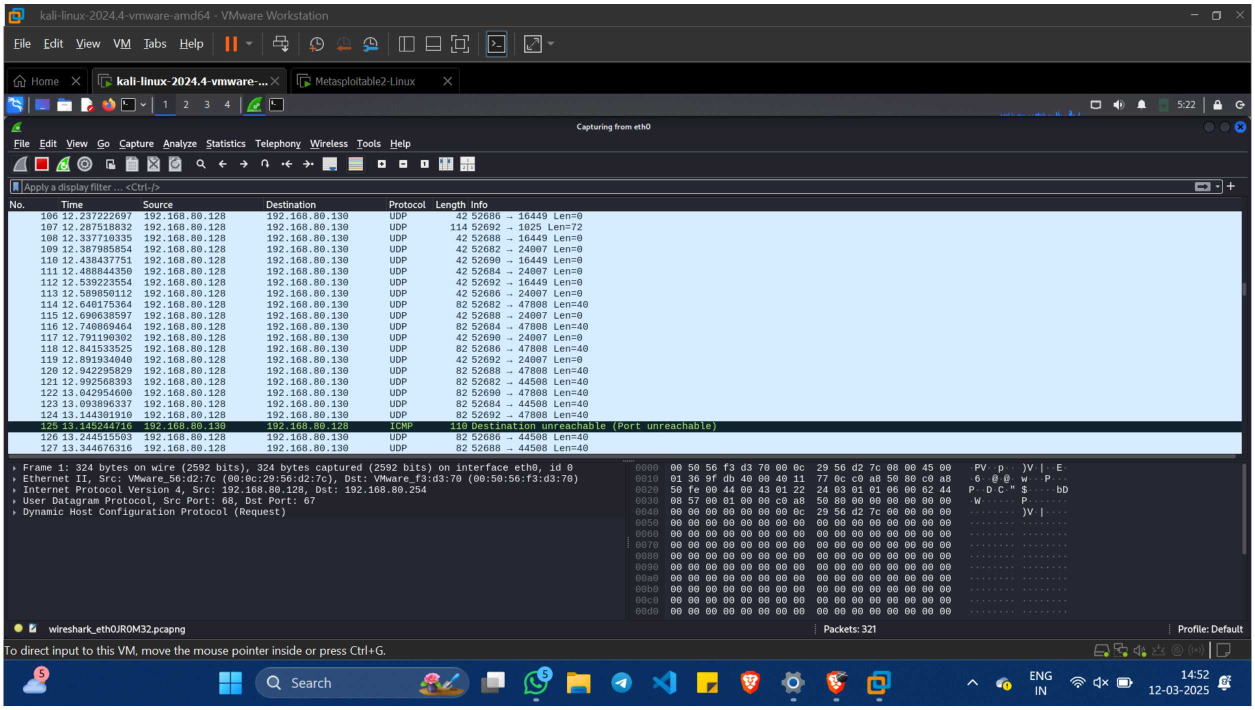

Wireshark Analysis

Wireshark was used to monitor traffic.

- Observed TCP Handshake Attempts
    - SYN packets were sent.
    - RST (Reset) responses meant closed ports.
    - The open port 111 completed the three-way handshake.

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